Lecture 11: Inequalities

Problem 294 Prove that

$$\frac{1}{2} < \frac{1}{101} + \frac{1}{102} + \cdots + \frac{1}{200} < 1$$

Solution For any $n = 101, ..., 200$, we have

$$\frac{1}{200} \leq \frac{1}{n} < \frac{1}{100}.$$

Recall that if $a > b$ and $c > d$ then $a + c > b + d$. So adding the inequalities together we have

$$\frac{1}{2} = 100 * \frac{1}{200} \leq \frac{1}{101} + \frac{1}{102} + \cdots + \frac{1}{200} < 100 * \frac{1}{100} = 1.$$

Problem 295 Prove that

$$\frac{1}{2} < 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{199} - \frac{1}{200} < 1$$

Solution Adding parathesis in a nice way!

$$(1 - \frac{1}{2}) + (\frac{1}{3} - \frac{1}{4}) + \cdots + (\frac{1}{199} - \frac{1}{200}) > \frac{1}{2}$$

because the first term is $\frac{1}{2}$ and the rest are positive.

$$1 + (-\frac{1}{2} + \frac{1}{3}) + \cdots + (-\frac{1}{198} + \frac{1}{199}) - \frac{1}{200} < 1$$

because the first term is 1 and the rest are negative.

Problem 295+

$$\frac{1}{101} + \frac{1}{102} + \cdots + \frac{1}{200} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{199} - \frac{1}{200}$$

Solution

$$LHS = 1 + \frac{1}{2} + \cdots + \frac{1}{200} - (1 + \frac{1}{2} + \cdots + \frac{1}{100})$$

$$RHS = 1 + \frac{1}{2} + \cdots + \frac{1}{200} - 2(\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{200})$$

So they are the same. This gives the general formula

$$\frac{1}{n+1}+\frac{1}{n+2}+\cdots+\frac{1}{2n}=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots+\frac{1}{2n-1}-\frac{1}{2n}$$

Problem 296+ for any positive integer n and positive number x, we have

$$(1+x)^n>1+nx.$$

In particular $(1.01)^{100}>1+1=2$.

Solution By the binomial expansion formula,

$$(1+x)^n=1+\binom{n}{1}x+\binom{n}{2}x^2+\cdots+\binom{n}{n}x^n>1+nx.$$

Problem 297+

$$1+\frac{1}{2^2}+\frac{1}{3^2}+\cdots+\frac{1}{100^2}<2$$

Solution We have $\frac{1}{n^2}<\frac{1}{n(n-1)}$.

$$\begin{aligned} LHS &< 1+\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots+\frac{1}{99\cdot 100} \\ &= 1+1-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\cdots+\frac{1}{99}-\frac{1}{100}<2 \end{aligned}$$

In fact, this sum goes to infinity is the famous Bassel's problem.

Theorem (Euler)

$$1+\frac{1}{2^2}+\frac{1}{3^2}+\cdots+=\frac{\pi^2}{6}$$

Problem 300

$$1+\frac{1}{2}+\frac{1}{3}+\cdots+=+\infty$$

Solution

$$\begin{aligned} &1+\frac{1}{2}+(\frac{1}{3}+\frac{1}{4})+(\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8})+\ldots. \\ &<1+\frac{1}{2}+\frac{1}{2}+\frac{1}{2}+\ldots.=+\infty \end{aligned}$$

Arithmetic and Geometric Means

For two nonnegative numbers a and b, we define

Arithmetic mean: $\frac{a+b}{2}$

Geometric mean: $\sqrt[2]{ab}$

Theorem

$$\sqrt[2]{ab} \leq \frac{a+b}{2}$$

and the equality holds if and only if $a = b$.

Proof 1: taking the square, we get

$$4ab \leq (a+b)^2$$

This is true because $(a+b)^2 - 4ab = (a-b)^2 \geq 0$.

geometric proof Draw a square with length $a+b$. Then you can embed $4ab$ as four retangles inside.

Theorem Given $a, b, c \geq 0$, we have

$$\sqrt[3]{abc} \leq \frac{a+b+c}{3}$$

Proof Let $p = \sqrt[3]{a}$, $q = \sqrt[3]{b}$, $r = \sqrt[3]{c}$. Then it suffices to show that

$$pqr \leq \frac{p^3+q^3+r^3}{3}$$

Notice that

$$p^3 + q^3 + r^3 - 3pqr = (p+q+r)(p^2+q^2+r^2-pq-qr-rp)$$

$$= \frac{1}{2}(p+q+r)((p-q)^2+(q-r)^2+(r-p)^2) \geq 0$$

So it is proved.